

The Practitioner's Chemistry

The actives that work, the chemical services that transform, and the caution every strand deserves

Learning Objectives

- Name the haircare actives with genuine evidence behind them, and what each actually does
- Explain, at a working level, how relaxers, perms, and permanent colour each alter the strand
- Identify the chemical service combinations that create the highest risk of damage
- Explain why patch and strand testing are non-negotiable before any chemical service

Ingredients versus marketing

The haircare aisle is one of the most confidently marketed categories in beauty, and much of that confidence outpaces the evidence. This module is not a list of everything on a label. It is a shorter, more honest list — the actives that have real support behind them, understood well enough that you can explain to a client, in plain language, what they are actually paying for.

The core actives

- **Proteins (keratin, hydrolysed silk, wheat protein)** — temporarily patch gaps in a damaged cuticle, improving strength and reducing breakage. Overuse causes stiffness and increased breakage, which is why protein and moisture must always be balanced.
- **Panthenol (pro-vitamin B5)** — draws and holds moisture in the strand, improving elasticity and shine measurably in clinical testing.
- **Silicones (dimethicone and relatives)** — coat the cuticle to smooth and add shine instantly. Not inherently harmful, contrary to a persistent myth, but water-insoluble

silicones can build up over time and require a clarifying shampoo to remove.

- **Biotin and other B-vitamins** — genuinely necessary for healthy hair production, but only when a true deficiency exists; supplementing an already-sufficient diet shows little added benefit, despite heavy marketing.
- **Minoxidil** — the one topical ingredient with strong clinical evidence for regrowth in androgenetic alopecia. It is a medical treatment, not a cosmetic one, and belongs to the referral conversation from Module Three, not to a retail shelf recommendation made casually.

◆ Figure 4.1 · The Core Actives



The core actives worth knowing well — what each does, and what it does not.

Chemical services: what actually happens to the strand

Return to Module One's disulfide bonds. Every service below works by the same basic principle — break the bonds, reshape the strand, then reform the bonds in the new shape.

- **Chemical relaxers (lye and no-lye).** A strong alkaline agent swells the cuticle and breaks disulfide bonds, allowing the hair to be combed straight before a neutralising step reforms the bonds in that straightened shape. Lye relaxers act faster and are typically applied by a professional; no-lye formulas are gentler on the scalp but can be drier on the strand. This is a chemical service with genuinely high risk of scalp burns and structural damage if timing or application is wrong.
- **Perms.** The reverse process — thioglycolate-based solutions break disulfide bonds so the hair can be wound around a rod, then a neutraliser reforms the bonds around the curl shape.

- **Permanent hair colour.** Ammonia or an ammonia alternative swells the cuticle open, hydrogen peroxide lightens the natural pigment in the cortex, and dye molecules deposit and lock into the cortex as the cuticle closes back down. This is why permanent colour changes texture slightly even when the client only wanted a change of shade.
- **Bond-building treatments.** A newer category of additive designed to protect and reinforce disulfide bonds during a chemical service, reducing (though never eliminating) the damage of colour or lightening.

Where the highest risk lives

The most damaging moment in most salons is not any single service — it is the combination of services performed too close together on hair that has not had time to recover. Chemically relaxing hair that was coloured within the past two weeks, or lightening hair that has already been relaxed, stacks multiple disulfide-breaking processes on a strand that may not have the structural integrity to survive a second chemical event. This is why a strand test — cutting a small section, applying the intended service, and observing the result before committing the whole head — is never optional on chemically compromised hair, and why an honest "not today" is sometimes the most professional thing you can say to a client eager for a dramatic change.

Patch and strand testing, without exception

Every chemical service begins with a patch test — a small amount of product applied behind the ear or inner elbow, 24–48 hours before the full service, checked for redness, swelling, or irritation. This is not a suggestion; permanent colour and relaxer formulas can trigger genuine allergic reactions, occasionally severe ones, and a patch test is the only way to know in advance. Document every patch test on the client's consultation card, with the date and result.

Key Terms

0

[object Object]

1

[object Object]

2

[object Object]

3

[object Object]

4

[object Object]

Practical Activity · The Ingredient Audit

Bring in the ingredient list of one haircare product you use personally. Identify every active ingredient discussed in this module, and note any claims on the packaging that this module's evidence does not support. **Bring your findings to discuss in your next live session.**